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Course: Data Structures and Algorithms

Trees

Evis Plaku

Why trees?

When your data structures have more branches than your family tree...

Modern family tv show

Introduction to trees

Why trees?

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Trees organize data in a **hierarchical structure**, resembling branching systems in nature

Benefits

- **Efficient searching**: fast data retrieval (e.g., BST)
- **Hierarchical representation**: ideal for modeling relationships (e.g., file systems)
- **Flexibility**: easy insertion, deletion, and updates
- **Optimized traversals**: efficient ways to visit all nodes

Use cases

- **File systems**: organize files and directories and **databases**: fast indexing and searching
- **Decision trees**: machine learning classification
- **Expression trees**: represent mathematical expressions
- **Game trees**: decision-making in games

Introduction to trees

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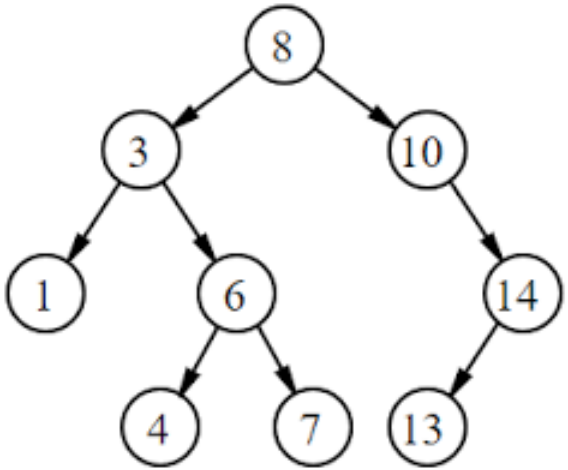
A hierarchical data structure consisting of nodes connected by edges

- A tree is a way to organize data in a hierarchy
- It has a **root** node, which is the starting point
- Each node can have **child nodes** that branch out
- No cycles: you can't go back to the same node by following the branches



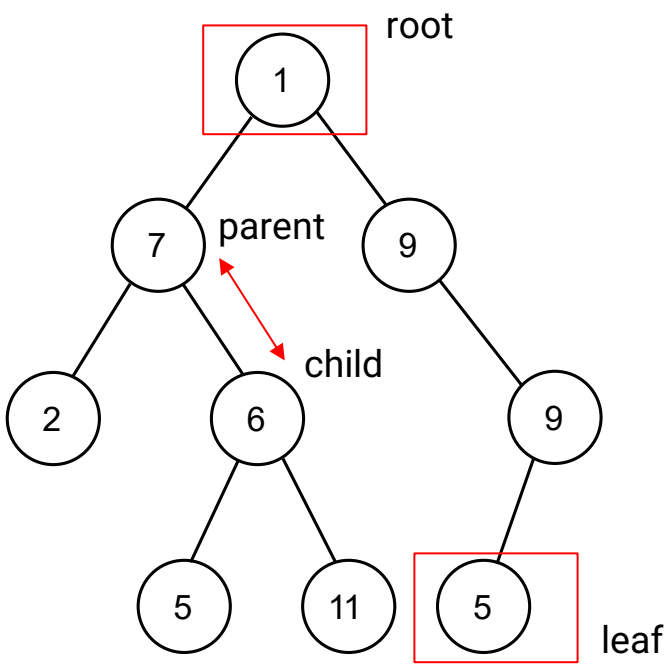
Terminology of trees

- Node: a basic unit of a tree that contains data

- Root: the topmost node in the tree, where the hierarchy starts

- Leaf: a node with no children, located at the "end" of branches

- Parent: a node that has one or more child nodes. Child: a node directly connected to another node above it (the parent)



Terminology of trees

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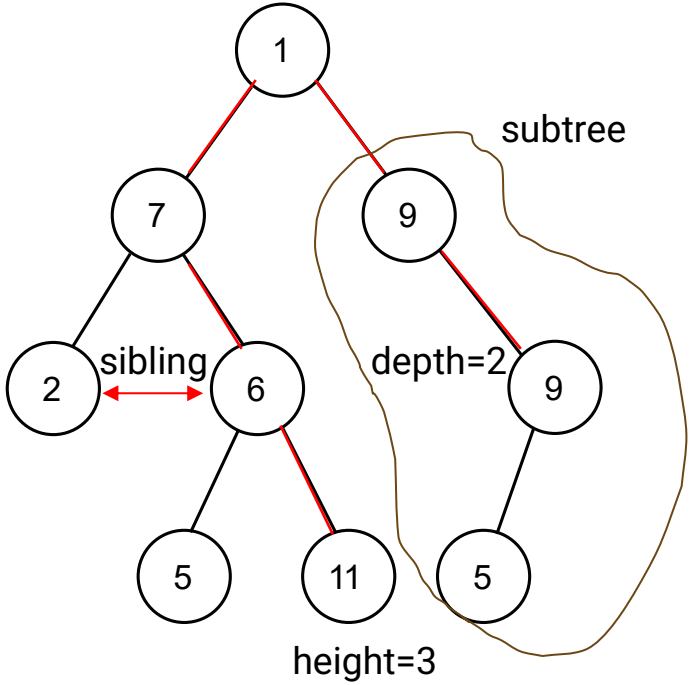
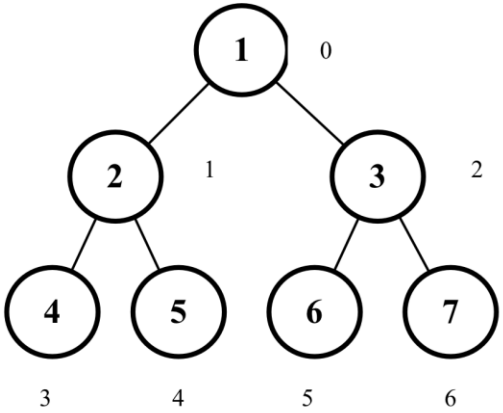
- Sibling: nodes that share the same parent
- Depth: the number of edges from the root to a node
- Height: the number of edges from a node to its farthest leaf
- Level: a node's level is distance from the root
- Subtree: a tree formed by a node and all its descendants

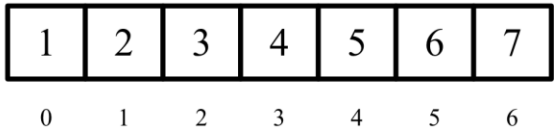


Array representation of trees

- The root is at index 0
- For any node at index i:
  - Left child is at index 2i + 1
  - Right child is at index 2i+ 2



Trees can be represented using an **array**



Array representation of trees

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- The `ArrayTree` initializes an array `tree` to store the nodes
- Methods `setValue` and `getValue` are used to respectively set and retreive node values
- Methods `getLeftChild` and `getRightChild` are used to respectively get the indices of the left and right child

```java
public class ArrayTree {

    private int[] tree;  // Array to hold tree nodes
    private int size;    // Size of the tree

    public ArrayTree(int size) {
        this.size = size;
        this.tree = new int[size];
    }

    public void setValue(int index, int value) {
        // TODO: Add logic to set value at the given index
    }

    public int getValue(int index) {
        // TODO: Add logic to return the value at the given index
        return -1;  // Placeholder
    }

    public int getLeftChild(int index) {
        // TODO: Add logic to calculate left child index
        return -1;  // Placeholder
    }

    public int getRightChild(int index) {
        // TODO: Add logic to calculate right child index
        return -1;  // Placeholder
    }
}
```

Linked list representation of trees

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- Each node in the tree is an object that contains:
  - **Data**: the value stored in the node
  - **Left child pointer**: a reference to the left child node
  - **Right child pointer**: a reference to the right child node

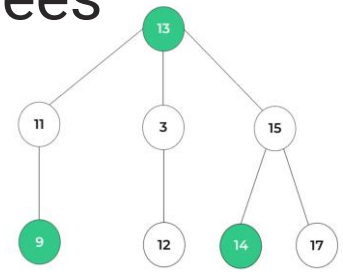
```java
public class TreeNode {

    int data;
    TreeNode leftChild;
    TreeNode rightChild;

    public TreeNode(int data) {
        this.data = data;
        this.leftChild = null;
        this.rightChild = null;
    }
}
```

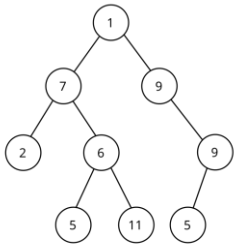
No predefined size, unlike array representation.

The tree grows and shrinks as needed.

Easy to insert, delete, and manipulate nodes

Types of trees





General trees

- **A tree where each node can have any number of children**
- No restriction on the number of children a node can have
- Used for more complex hierarchical structures

Binary trees

- **A tree where each node has at most two children**
- Each node has two child pointers (left and right)
- Common in computer science due to simplicity and efficiency in operations

Binary Search Trees (BST)

Binary search trees: properties and applications

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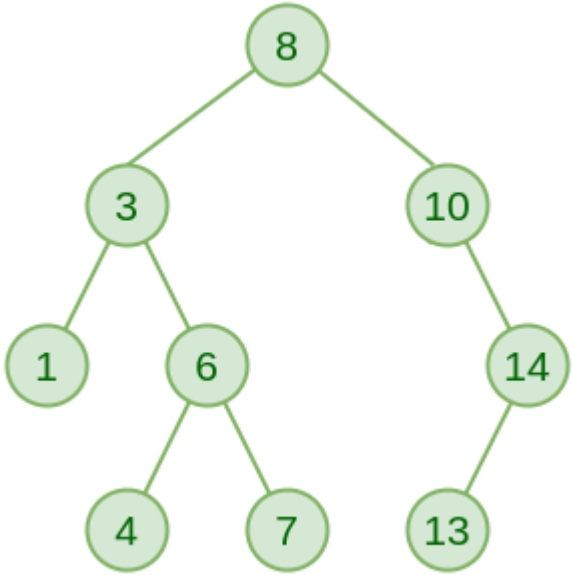
- Binary Search Tree (BST): a binary tree where for every node
  - The left child's value is less than the node's value
  - The right child's value is greater than the node's value
- Searching for a value has an average time complexity of $O(\log n)$ if the tree is balanced
- All nodes follow the $left < parent < right$ rule



Applications of binary search trees

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Database indexing

File system directories

Online ticketing systems

Game leaderboards

Navigation systems

Online shopping filters

Full trees and perfect trees

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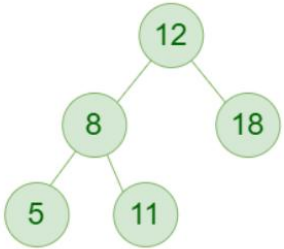
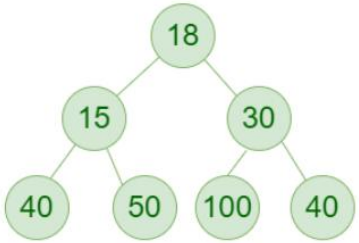
Full binary tree

- A binary tree where every node has either 0 or 2 children
- All internal nodes (non-leaf nodes) must have exactly two children
- Leaf nodes have no children



Perfect binary tree

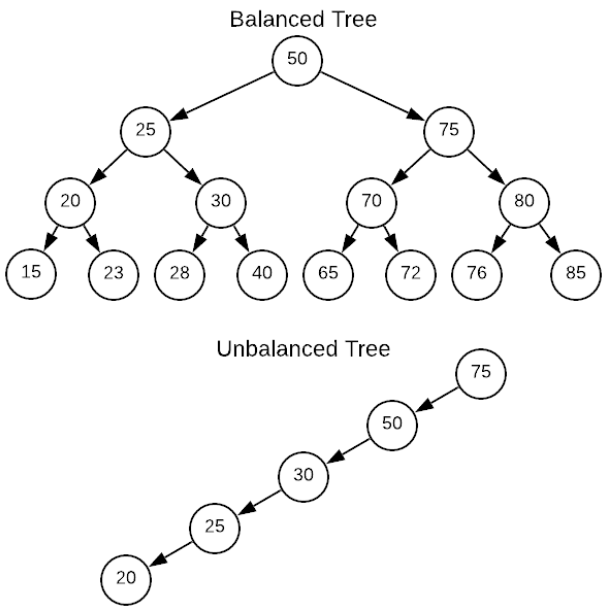
- A binary tree where all leaf nodes are at the same level, and every internal node has exactly two children
- All levels are fully populated
- The height of the tree is well-defined



Balanced binary search trees

A Binary Search Tree is balanced when the height difference between left and right subtrees is at most 1

- Ensures O(log n) time complexity for operations
- Prevents degeneration into a linear structure
- Maintains optimal memory and space usage
- Supports efficient algorithms



Key operations in BSTs: insertion

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- Start at the root
- Compare the value with the current node
- If the value is less, move to the left child; if greater, move to the right child
- Repeat until you find an empty spot (null) where the new node will be inserted

$O(h)$ where h is the height of the tree

- Best case: $O(\log n)$ for balanced tree
- Worst case: $O(n)$ for skewed tree

Key operations in BSTs: searching

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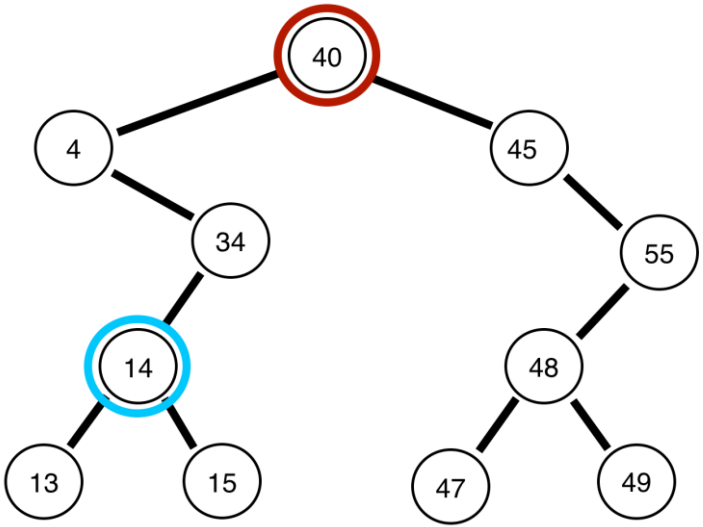
- Start at the root
- Compare the target value with the current node's value
- If less, move to the left child; if greater, move to the right child
- Repeat until the target is found or you reach a null (value not in tree)



$O(h)$ where h is the height of the tree

- Best case: $O(\log n)$ for balanced tree
- Worst case: $O(n)$ for skewed tree

Key operations in BSTs: deletion

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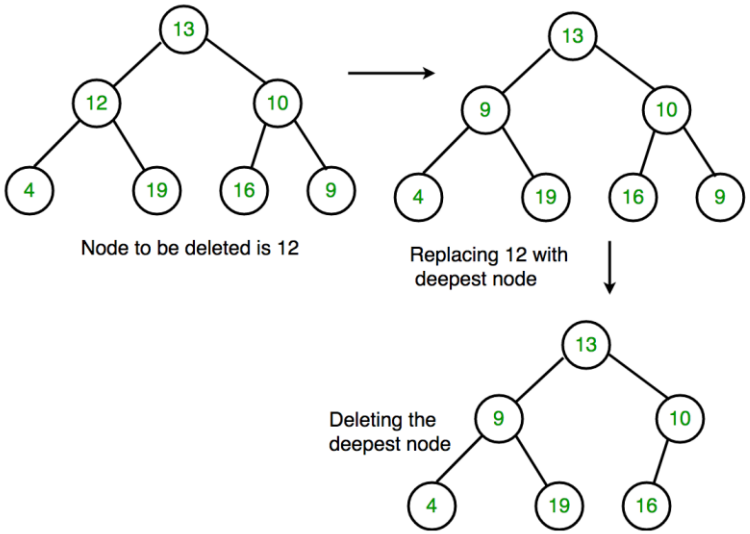
- Case 1: node has no children (leaf): simply remove the node
- Case 2: node has one child: replace the node with its child
- Case 3: node has two children: find the in-order successor (or predecessor), replace the node with it, and remove the successor



$O(h)$ where h is the height of the tree

- Best case: $O(\log n)$ for balanced tree
- Worst case: $O(n)$ for skewed tree

Traversals in Binary Search Trees

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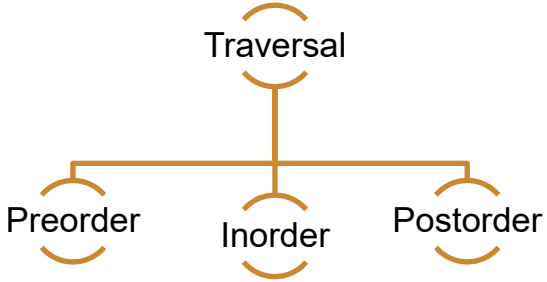
Traversal means visiting all nodes in a tree in a systematic way

- Traversals allow us to search, print, or manipulate every node in a desired order
- Traversals help visualize and debug tree structure during operations like insertions or deletions
- Traversals define how data is saved to or read from a tree (e.g., saving a tree to a file)



Order matters!

Preorder traversal (Node → Left → Right)

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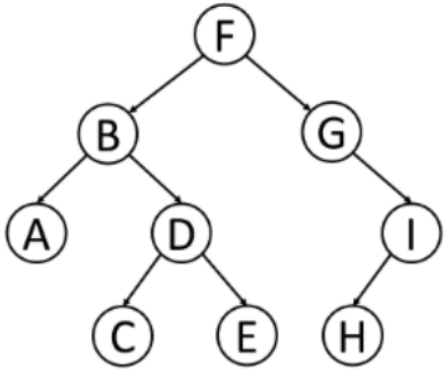
Order

1. Visit the current node
2. Traverse the left subtree
3. Traverse the right subtree



Preorder:

- **Copying or cloning trees** (ensures the structure is preserved starting from the root)
- **Serializing trees** (preorder is used to flatten trees into strings or arrays for storage and reconstruction)
- **Evaluating prefix expressions** (preorder visits operators before operands)

Inorder traversal (Left → Node → Right)

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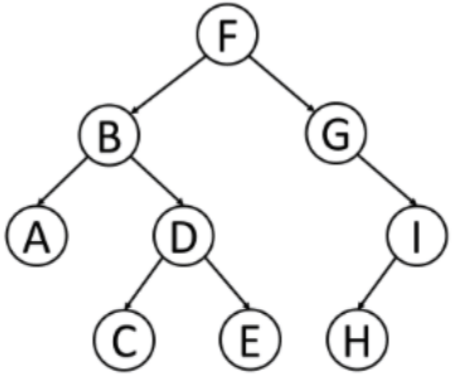
Order of traversal

1. Traverse the left subtree
2. Visit the current node
3. Traverse the right subtree



Inorder:

- **Retrieving sorted data** (used to extract elements from a BST in non-decreasing order)
- **BST validation** (if inorder traversal yields a non-sorted list, the tree violates BST properties)
- **Infix expressions evaluation** (especially in expression trees for mathematical expressions)

Postorder traversal (Left → Right → Node)

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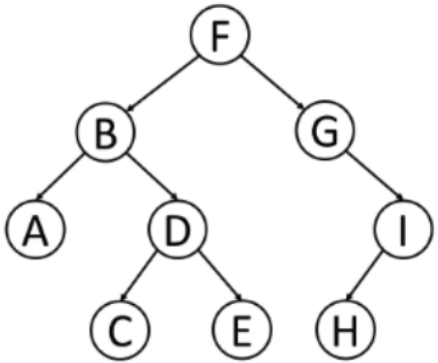
Order of traversal

1. Traverse the left subtree
2. Traverse the right subtree
3. Visit the current node



Postorder:

- **Tree deletion** (safely deletes child nodes before the parent)
- **Recursive tree operations** (great for structural modifications like pruning, transformations)
- **Postfix expressions evaluation** (in expression trees, postorder supports evaluating mathematical expressions)

Example of traversal in real-world use cases

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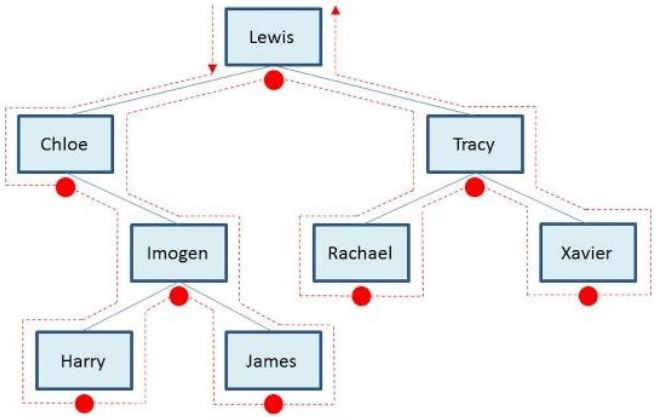
- Assume a BST is storing `Student` or `Employee` data, sorted by `id`

- **Inorder** traversal is useful retrieving for all records in ascending order (generate reports, ranked lists etc)

- **Preorder** traversal is useful to serialize or transfer data (export the tree to a file)



- **Postorder** traversal is useful to remove records (e.g., delete students graduation this year)

Operations complexity in BST

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Operation	Balanced BST	Unbalanced BST
Insert	$O(\log n)$	$O(n)$
Delete	$O(\log n)$	$O(n)$
Search	$O(\log n)$	$O(n)$
Traversal	$O(n)$	$O(n)$

Balanced BSTs ensure efficient operations with $O(\log n)$ time

Unbalanced BSTs degrade to $O(n)$ time in worst cases

Key Takeaways

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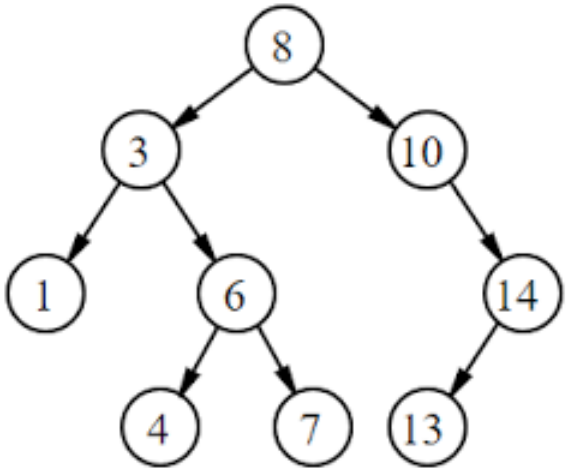
Trees are hierarchical structures ideal for representing nested or ordered data

- Binary Search Trees (BSTs) maintain a sorted structure: `Left < Root < Right`
- BST operations like search, insert, and delete are efficient when the tree is balanced
- BSTs are widely used in databases, indexing, and building efficient search systems



Helpful resources on recursion

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- 🔍 VisualAlgo to visualize BST tree and operations
- 🎥 Software testing tutorial on Binary Search Trees in Java
- 📚 Implementing a Binary Search Tree in Java
- 💻 LeetCode set of problems on trees

Quote of the Week

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In a tree, roots come first

just like in life